

Quiz 1: DIC on Discrete Structures

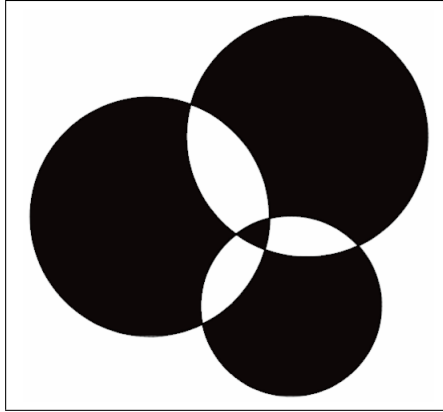
25 marks, 50min

28 Aug 2024

Instructions:

- Attempt *all* questions. Write all answers and proofs carefully. If you are *making any assumptions or using results proved in class, state them clearly*.
- All sets considered below are general sets that can be infinite. Hence, you must not assume that they are finite or countable, unless clearly specified otherwise.
- In this course, one of our aims is to learn how to write good proofs, hence considerable weightage will be given to *clarity and completeness* of proofs.
- Do not copy or use any other unfair means. Offenders will be reported to the Disciplinary Action Committee.
- As usual, $\mathbb{N}$ stands for the set of natural numbers and $\mathbb{Z}$ stands for the set of integers. A, B, U are arbitrary sets unless otherwise stated.

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1. (9 marks) True or false. If true, give a short proof, if false give a counter-example and a short justification why its a counter-example.
 - (a) For all $x, y \in \mathbb{N}$, if $(x \cdot y^2)$ is odd, then $(5x + 3y)$ is odd.
 - (b) Let A, B be two infinite sets. If A is countable and $B \subseteq A$ then B must be countable.
 - (c) For sets $A, B \subseteq U$ if there is a bijection from A to B then there is a bijection from complement of A , i.e., $U \setminus A$ to complement of B , i.e., $U \setminus B$.
 2. (5 marks) Consider the statement:
Every integer greater than 7 is the sum of a nonnegative integer multiple of 3 and a nonnegative integer multiple of 5.
Write this as a proposition using quantifiers and prove it formally by using (strong) induction or WOP.
 3. (6 marks) Which of the following sets are countable? Justify your answer. If uncountable, use Cantor's diagonalization to show that this is the case.
 - (a) The set of all functions from $\{0, 1\}$ to $\mathbb{Z}$.
 - (b) The set of all functions from $\mathbb{Z}$ to $\{0, 1\}$.
 4. (5 marks) n circles (where $n \geq 1$) are given on the plane, dividing it into several regions. Show, by induction, that one can color the plane with two colors, such that no two regions with a common boundary line have the same color. See Figure below for an example.



Note that having a single point of contact between two regions does not count as a common boundary line.

5. (0 marks) In your opinion, who was more awesome: Cantor or Russell? Why?